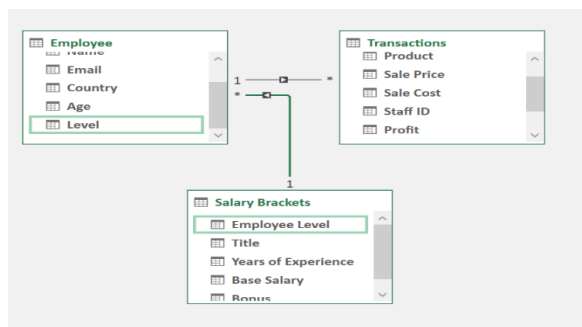


Project – Advanced Data Modeling and KPI Analysis with Power Pivot

This project involved building a data model using **Power Pivot**, creating relationships between tables, performing calculations using **DAX**, and conducting performance analysis using **KPIs (Key Performance Indicators)**. The key steps I followed are detailed below:

1. I opened a new Excel file and launched **Power Pivot > Manage** to access the Power Pivot data model interface. Since there was no existing data, I used **Get External Data > From Other Sources > Excel File** to import the data.
2. I imported all three worksheets from the selected Excel file into Power Pivot, ensuring that the first row was used as column headers.
3. Using the **Diagram View** in Power Pivot, I created **relationships** between the tables by linking:
 - a) Staff ID in the transaction table to Staff ID in the employee table.
 - b) Employee Level in the salary table to the corresponding field in the employee table.



4. After establishing the relationships, I returned to **Data View** and created a **calculated column** for Profit using formula and renamed the column accordingly.

[Profit] fx = [Sale Price]-[Sale Cost]					
Customer	Product	Sale Price	Sale Cost	St...	Profit
1 Aliah	Toyota	\$20,000.00	\$14,580.00	10011	\$5,420.00
2 Clara	Honda	\$22,550.00	\$18,040.00	10013	\$4,510.00
3 Mikaela	Honda	\$14,895.00	\$11,022.30	10013	\$3,872.70
4 Joshua	Mitsubishi	\$20,000.00	\$15,400.00	10011	\$4,600.00
5 Kevin	Nissan	\$18,000.00	\$14,400.00	10011	\$3,600.00
6 Rodrigo	Toyota	\$19,500.00	\$14,430.00	10011	\$5,070.00
7 Jack	Toyota	\$20,000.00	\$15,400.00	10011	\$4,600.00
8 Jackeline	Toyota	\$22,000.00	\$9,900.00	10012	\$12,100.00
9 Lino	Toyota	\$25,000.00	\$13,750.00	10014	\$11,250.00
10 Jess	Toyota	\$18,000.00	\$9,900.00	10015	\$8,100.00
11 Kayla	Nissan	\$16,580.00	\$10,777.00	10012	\$5,803.00
12 Cris	Honda	\$11,999.00	\$7,799.35	10012	\$4,199.65
13 Namto	Mitsubishi	\$22,500.00	\$12,375.00	10012	\$10,125.00
14 Take	Subaru	\$23,500.00	\$15,275.00	10015	\$8,225.00
15 Shintoro	Toyota	\$23,000.00	\$12,650.00	10014	\$10,350.00

5. To retrieve the **Employee Name** from the related table, I used the **RELATED** function in a new column, which dynamically fetched employee names based on the linked Staff ID.

[Empl... fx = RELATED(Employee[Name])						
Customer	Product	Sale Price	Sale Cost	St...	Profit	Employee Name
1 Aliah	Toyota	\$20,000.00	\$14,580.00	10011	\$5,420.00	Sarah
2 Clara	Honda	\$22,550.00	\$18,040.00	10013	\$4,510.00	Jones
3 Mikaela	Honda	\$14,895.00	\$11,022.30	10013	\$3,872.70	Jones
4 Joshua	Mitsubishi	\$20,000.00	\$15,400.00	10011	\$4,600.00	Sarah
5 Kevin	Nissan	\$18,000.00	\$14,400.00	10011	\$3,600.00	Sarah
6 Rodrigo	Toyota	\$19,500.00	\$14,430.00	10011	\$5,070.00	Sarah
7 Jack	Toyota	\$20,000.00	\$15,400.00	10011	\$4,600.00	Sarah
8 Jackeline	Toyota	\$22,000.00	\$9,900.00	10012	\$12,100.00	Alex
9 Lino	Toyota	\$25,000.00	\$13,750.00	10014	\$11,250.00	Mike
10 Jess	Toyota	\$18,000.00	\$9,900.00	10015	\$8,100.00	Ana
11 Kayla	Nissan	\$16,580.00	\$10,777.00	10012	\$5,803.00	Alex
12 Cris	Honda	\$11,999.00	\$7,799.35	10012	\$4,199.65	Alex
13 Namto	Mitsubishi	\$22,500.00	\$12,375.00	10012	\$10,125.00	Alex
14 Take	Subaru	\$23,500.00	\$15,275.00	10015	\$8,225.00	Ana
15 Shintoro	Toyota	\$23,000.00	\$12,650.00	10014	\$10,350.00	Mike

6. Next, I created an **explicit DAX measure** for **Total Profit**. This measure was stored in the model and later used for analysis and KPI creation.

[Profit] fx Total Profit:=sum([Profit])						
	Customer	Product	Sale Price	Sale Cost	St...	Profit
1	Aliah	Toyota	\$20,000.00	\$14,580.00	10011	\$5,420.00
2	Clara	Honda	\$22,550.00	\$18,040.00	10013	\$4,510.00
3	Mikaela	Honda	\$14,895.00	\$11,022.30	10013	\$3,872.70
4	Joshua	Mitsubishi	\$20,000.00	\$15,400.00	10011	\$4,600.00
5	Kevin	Nissan	\$18,000.00	\$14,400.00	10011	\$3,600.00
6	Rodrigo	Toyota	\$19,500.00	\$14,430.00	10011	\$5,070.00
7	Jack	Toyota	\$20,000.00	\$15,400.00	10011	\$4,600.00
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13	Namto	Mitsubishi	\$22,500.00	\$12,375.00	10012	\$10,125.00
14	Take	Subaru	\$23,500.00	\$15,275.00	10015	\$8,225.00
15	Shintoro	Toyota	\$23,000.00	\$12,650.00	10014	\$10,350.00
						Total Profit: 155020.4

7. I inserted a **Pivot Table** based on the Power Pivot data model and demonstrated how to pull fields from multiple tables using **Employee Name** from the employee table and **Sum of Profit** from the transactions table to analyze each employee's performance. Additional analysis was performed by adding **Employee Level** to the pivot table layout, revealing insights into performance versus seniority (e.g., high-performing junior employees).

Row Labels	Level	Sum of Profit
Mike	4	\$48,280.75
Alex	2	\$35,802.45
Sarah	1	\$29,829.50
Ana	3	\$22,125.00
Jones	3	\$13,182.70
Sam	1	\$5,800.00

8. Finally, I introduced **Key Performance Indicators (KPIs)**. I created a new KPI based on the Total Profit measure, setting a **target threshold** (e.g., \$20,000) and choosing an appropriate visual indicator. The KPI status was added to the pivot table, clearly highlighting which employees met or fell short of the performance goal.

Row Labels	Level	Sum of Profit	Total Profit
Mike	4	\$48,280.75	🟢
Alex	2	\$35,802.45	🟢
Sarah	1	\$29,829.50	🟢
Ana	3	\$22,125.00	🟢
Jones	3	\$13,182.70	🟡
Sam	1	\$5,800.00	🔴